

# Generative Retrieval for Unsupervised Text-Based Person Search

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**Abstract**—Text-based person search (TBPS) aims to retrieve images of a target person from a large image gallery based on a given natural language description. Most existing methods rely on supervised learning with manually annotated image-text pairs. In this paper, we explore unsupervised TBPS, with only unlabeled images. We propose GTR<sup>+</sup>, a two-stage generation-then-retrieval framework. In the generation stage, we introduce a tiered description generation framework designed to produce fine-grained and stylistically diverse textual descriptions through a three-tier sequential process. The base tier leverages an automated question-and-answer mechanism to generate basic visual attribute descriptions; the intermediate tier enhances fine-grained detail using an inter-sample contrastive mechanism; the advanced tier further enriches textual diversity via a stylized expansion mechanism. In the retrieval stage, to mitigate the impact of noisy pseudo texts, we develop an adaptive confidence-weighted retrieval learning framework. We model image-text pairs as clean or noisy using a Gaussian Mixture Model, calibrated by real-time image-text similarity and static text generation probability from the prior stage, yielding adaptive sample weights during training. Beyond that, we also contribute LargeFine-Person, a large-scale TBPS dataset with high-quality, fine-grained, and diverse textual annotations, enabling a practical and generalizable TBPS pre-training benchmark under unsupervised setting. Experiments on multiple TBPS benchmarks demonstrate the effectiveness and generalization of both GTR<sup>+</sup> and LargeFine-Person. Code is available at: <https://github.com/Flame-Chasers/GTR>.

**Index Terms**—Text-based person search; unsupervised learning, image captioning, person re-identification.

## 1 INTRODUCTION

Text-based person search (TBPS) [1], [2], [3] involves retrieving images of a specific individual from an image gallery based on textual descriptions. This task shares similarities with classical person re-identification [4], [5] and image-text retrieval tasks [6], [7], yet presents unique challenges of its own. In contrast to person re-identification that relies on image-based queries, TBPS offers a more accessible solution with free-form text as queries, but faces challenges in cross-modal learning due to differences between visual and textual data. Compared to general image-text retrieval, TBPS focuses on identifying individuals, which requires attention to fine-grained details and presents greater challenges due to increased intra-class variability (e.g., varying appearances of the same individual) and reduced inter-class variability (e.g., visually similar individuals). Despite these challenges, TBPS has gained attention in recent years for its potential in real-world applications, such as surveillance for locating suspects or missing persons.

The current mainstream research in TBPS [8], [9], [10] focuses on developing robust modality-invariant representations through sophisticated cross-modal alignment frameworks. While these approaches improve performance, they

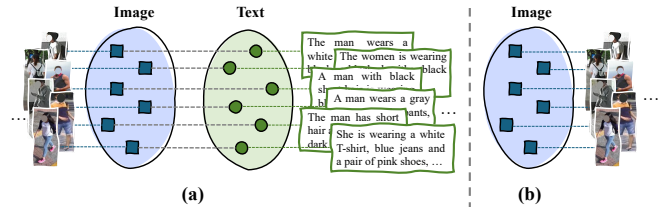


Fig. 1: Illustration of (a) canonical TBPS with supervised image-text data, (b) TBPS rely solely on image corpora.

heavily depend on supervised image-text data (i.e., person images with manually annotated descriptions) for training, as shown in Fig. 1 (a). This reliance presents challenges for real-world applications, where annotating large-scale person images with textual descriptions is labor-intensive and time-consuming. Therefore, this work focuses on unsupervised TBPS, eliminating the need for annotated text corpora and relying solely on image corpora for effective retrieval, as shown in Fig. 1 (b). For this, we present a basic two-stage solution, generation-then-retrieval (GTR<sup>+</sup>), which firstly generates pseudo text descriptions corresponding to the person images for remedying the absent annotation, and then trains a retrieval modal in a supervised manner.

**Generation.** Given the person images, our goal is to generate their paired pseudo textual descriptions that accurately depict the fine-grained details with diverse sentence structures. A simple approach is to use pre-trained Vision-Language Models (VLMs) [11] for direct image captioning. However, as shown in Fig. 2 (a), these models often fail to produce satisfactory results due to their pre-training on

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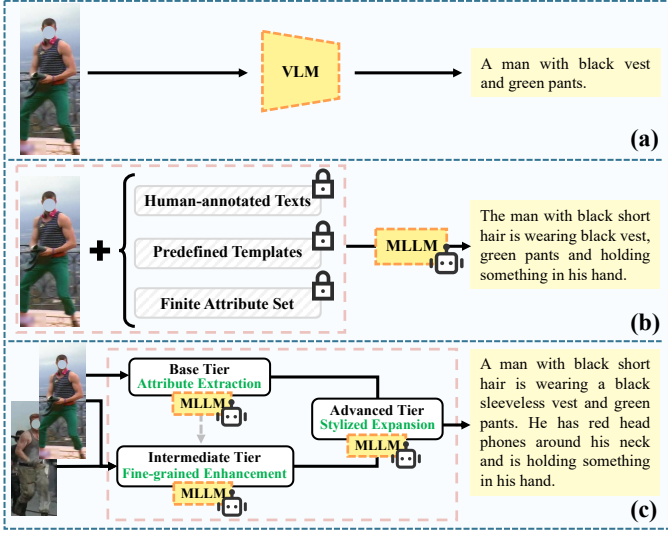


Fig. 2: Illustration of the text generation for a given person image. (a) Direct image captioning using a pre-trained vision-language model. (b) Description generation via multimodal large language model with auxiliary information. (c) The proposed tiered description generation framework.

general image-text pairs, limiting their ability to describe the nuanced features specific to person images. Some studies [12], [13], [14], [15] leverage Multimodal Large Language Models (MLLMs) for fine-grained description generation. However, these methods often rely on predefined templates, finite attribute set, or annotated texts as guidance. Consequently, the generated textual descriptions provide limited information, restricting both the richness and granularity of the outputs, as shown in Fig. 2 (b). To address these limitations, we propose a tiered description generation framework which constructs fine-grained and diverse textual descriptions through a self-contained, multi-tier process (Fig. 2 (c)). Specifically, it consists of three sequential tiers. 1) Base tier. Instead of relying on fixed templates, we design an automated question-and-answer mechanism that enables MLLM to self-query and extract fine-grained visual attributes (e.g., gender, clothing type, clothing color, accessories). This process aims to generate a basic textual description that captures the essential appearance characteristics of the person. 2) Intermediate tier. We develop an inter-sample contrastive mechanism to further capture subtle visual details. By comparing the target image with the hard-negative image, MLLM is prompted to highlight both similarities and differences, enhancing the granularity and informativeness of the descriptions. 3) Advanced tier. Building upon the results of the first two tiers, we propose a stylized expansion mechanism to enhance textual style diversity. The mechanism leverages MLLM to automatically enrich descriptions in various styles, guided by the previously generated texts. Overall, our framework produces fine-grained and stylistically diverse textual annotations in a tier-wise manner, which serve as effective supervision for training TBPS model.

**Retrieval.** This stage adopts generated image-text pairs to employ supervised training of the retrieval model. The retrieval model can, in principle, be integrated with any

existing TBPS method [3], [16]. However, it is important to note that these TBPS methods are typically designed to train on manually annotated and well-aligned image-text pairs. In contrast, the constructed pairs used as training data in our approach may contain noise due to potential hallucinations and inaccuracies introduced during the preceding generation stage. To mitigate the impact of noise on the retrieval model training, we propose an adaptive confidence-weighted retrieval learning framework. Confidence scores, reflecting image-text correspondence reliability, are adaptively incorporated into the loss to calibrate error propagation, assigning higher weights to more reliable pairs and enhancing their influence on learning. Confidence scores are computed using a Gaussian Mixture Model (GMM) that separates image-text pairs into clean and noisy components based on their similarity distribution. As training progresses, the similarity is updated adaptively—incorporating both real-time image-text alignment and prior text generation probability—enabling the GMM to iteratively refine sample reliability. This progressive refinement sharpens the clean-noisy distinction, yielding a more robust and accurate retrieval model.

In addition to proposing a GTR+ solution for unsupervised TBPS, we also contribute a large-scale TBPS dataset with fine-grained and diverse textual descriptions, named *LargeFine-Person*. The dataset is constructed by collecting and refining person images from LUPerson [17] and LPW [18], followed by generating corresponding textual descriptions using our proposed tiered description generation framework. As a result, we construct a total of 3.6 million image-text pairs. *LargeFine-Person* enables a practical and generalizable TBPS pre-training benchmark under unsupervised setting. A comparison between the proposed *LargeFine-Person* dataset and other pre-training TBPS datasets [13], [14], [15], [19], [20] is presented in Table 1. *LargeFine-Person* presents superior data quality—especially in terms of textual descriptions—attributed to the proposed tiered description generation framework.

In summary, our main contributions are as follows:

- 1) We propose a tiered description generation framework, integrating automated question-and-answer generation, inter-sample contrastive generation, and stylized expansion generation to produce semantically rich and diverse captions for person images. Based on this, we construct *LargeFine-Person*, a large-scale dataset of 3.6 million high-quality, fine-grained image-text pairs.
- 2) We propose an adaptive confidence-weighted retrieval learning framework that mitigates the impact of noisy pseudo-texts via self-adaptively weighting image-text pairs based on the confidence scores. These scores are iteratively refined using a Gaussian Mixture Model that integrates image-text similarity and text generation probability.
- 3) We conduct comprehensive experiments on multiple TBPS benchmarks. The results demonstrate the effectiveness and generalizability of our proposed GTR+ solution and *LargeFine-Person* dataset in unsupervised settings.

A preliminary version of this work has been presented

| Datasets                | Image Domain                |         |         | Text Domain |        |             |       |       |                   |
|-------------------------|-----------------------------|---------|---------|-------------|--------|-------------|-------|-------|-------------------|
|                         | Source                      | #Images | Denoise | Source      | #Texts | #Words/Text | #Adj. | #NCs. | Multi-granularity |
| MALS [20]               | Generation                  | 1.51M   | ✓       | Human.      | 1.51M  | 27          | 2.72  | 7.97  | ✗                 |
| LUPerson-T [19]         | LUPerson [17]               | 1.30M   | ✗       | Auto.       | 1.30M  | 27          | 4.39  | 5.86  | ✗                 |
| LUPerson-MLLM [13]      | LUPerson [17]               | 1.00M   | ✗       | Auto.       | 4.00M  | 29          | 3.47  | 8.62  | ✗                 |
| SYNTH-PEDES [14]        | LUPerson-NL [21] + LPW [18] | 4.79M   | ✓       | Human.      | 12.14M | 24          | 4.81  | 7.08  | ✗                 |
| HAM-PEDES [15]          | SYNTH-PEDES [14]            | 1.00M   | ✗       | Human.      | 2.00M  | 26          | 5.66  | 7.54  | ✗                 |
| LargeFine-Person (Ours) | LUPerson [17] + LPW [18]    | 1.20M   | ✓       | Auto.       | 3.60M  | 61          | 9.63  | 15.27 | ✓                 |

TABLE 1: Comparison between our pre-training dataset (LargeFine-Person) and existing pre-training datasets for TBPS. ‘Denoise’ indicates whether low-quality images are filtered out from the dataset. ‘Human.’ specifies that the generation of texts depends on manually annotated inputs, while ‘Auto.’ denotes that the text generation is fully automated without manual intervention. ‘#Words/Text’ represents the average number of words per text, reflecting the richness of information conveyed. ‘#Adj.’ denotes the average number of adjectives per text, reflecting the fine-grainedness of the descriptions. ‘#NCs.’ denotes the average number of noun chunks per text, indicating the completeness of the attribute coverage. ‘Multi-granularity’ indicates the presence of texts with varying levels of detail, reflecting the diversity of the textual data. Benefiting from the proposed tiered description generation framework, our dataset demonstrates advantages in both diversity and fine-grained detail.

at ACM MM 2023 [12]. This paper introduces the following significant advancements: (1) At the task level, while prior work predominantly focuses on TBPS without parallel image-text data—utilizing separately collected image and text corpora where textual data is still required, we focus on the more challenging and practical task of unsupervised TBPS, where only an image corpus is available. (2) At the model level, we propose a generation-then-retrieval framework, which extends and improves upon previous work in both text generation and retrieval modeling. In the text generation stage, we design a novel tiered description generation framework, which comprises three progressive components: an automated question-and-answer mechanism to capture basic attributes, an inter-sample contrastive mechanism to encode fine-grained details, and a stylized expansion mechanism to enrich the expressiveness of the generated descriptions. This tiered framework facilitates the generation of descriptions that are both semantically rich and stylistically diverse. It significantly improves upon previous work, which typically relied on manually designed prompts and external text corpora for image-to-attribute extraction and attribute-to-text conversion—often leading to monotonous outputs. In the retrieval learning stage, we introduce an adaptive confidence-weighted learning framework that enhances robustness to noisy data. Unlike previous work that used a static text generation probability as the confidence score, our method dynamically adjusts confidence scores during training, enabling them to adapt to the learning dynamics. This allows for more accurate weighting of training samples and better noise estimation, ultimately improving retrieval performance. (3) At the data level, leveraging the proposed tiered description generation framework, we additionally contribute a pre-training dataset LargeFine-Person, which enables a generalizable pre-training benchmark under unsupervised setting. These resources are expected to advance the application of TBPS in real-world scenarios.

## 2 RELATED WORK

### 2.1 Text-based person search

Existing TBPS approaches can be categorized into two main paradigms based on their primary focus: cross-modal

alignment and representation learning. The first line of research focuses on cross-modal alignment, aiming to map visual and textual features into a shared embedding space. Early methods relied on global feature alignment [22], [23], [24], while later works progressed to multi-granularity correspondences [16], [25], [26], [27], [28] and incorporated external tools—such as human parsing, pose estimation, and NLTK [29]—for fine-grained feature alignment [30], [31], [32]. Recent advances have achieved progress in self-adaptive semantic alignment [9], [33], [34], [35]. The second paradigm focuses on learning modality-invariant representations that generalize across modalities [36], [37], [37], [38], [39]. For instance, Wu *et al.* [40] introduced color-reasoning sub-tasks to enhance fine-grained cross-modal associations, while Wang *et al.* [41] proposed a color deprivation module to reduce over-reliance on color and learn more robust features. Recent efforts [2], [3], [38], [42], [43], [44], [45] have utilized large vision-language models like CLIP [46] and multimodal LLMs to enhance fine-grained retrieval. Several studies [10], [47] further incorporated multi-round interactions and contextual scene information [48] for better query-image alignment. However, these methods still rely heavily on costly, labor-intensive manually annotated image-text pairs, limiting their scalability and real-world applicability.

Recently, researchers have begun to recognize the annotation burden inherent in traditional TBPS and have explored less annotation-intensive paradigms. Several studies [49], [50], [51], [52], [53] focused on weakly supervised TBPS, where image-text pairs are available without explicit identity labels. Gao *et al.* [54] explored a semi-supervised setting with limited labeled pairs and abundant unlabeled images, while Jing *et al.* [55] studied cross-domain adaptation from labeled source domains to unlabeled targets. Though reducing annotation effort, these methods still require partial supervision. In contrast, fully unsupervised TBPS methods [1], [12], [56] required only unannotated images. They typically generated pseudo-text annotations followed by noise-robust learning. However, the performance is limited by: (1) template-based text generation yields limited and single-perspective descriptions, and (2) existing noise modeling strategies consider only local text-image similarity. In this work, we solve unsupervised TBPS

that addresses both limitations via three-tiered text generation and adaptive confidence-weighted retrieval learning.

## 2.2 Unsupervised Vision-Language Tasks

Considering that the paired image-text data are extremely labor-intensive to collect, significant efforts have been dedicated to advancing unsupervised learning in vision-language tasks, including unsupervised vision-language pretraining [57], [58], [59], [60], unsupervised text-to-image synthesis [61], and unsupervised image captioning [62], [63], [64], [65], [66]. In particular, unsupervised image captioning has seen flourishing development, driven by two major settings: unpaired image captioning (training with image-text data but unpaired each other) [66], [67], [68] and text-only image captioning (training exclusively on text corpora) [69], [70], [71]. For text-only image captioning, some studies [62], [69], [70] explored leveraging visual concepts to bridge the gap between textual and visual modalities, while others [72], [73], [74] proposed generating synthetic images from text data, training captioning models using these synthetic image-text pairs. Additionally, some research [71], [75], [76] focused on exploiting the cross-modal alignment capabilities of vision-language pretraining models by designing auxiliary modules to better adapt these models to downstream image captioning task. Nevertheless, these advancements in unsupervised vision-language tasks cannot be directly applied to unsupervised TBPS due to the significant differences in tasks. Conventional unsupervised vision-language tasks focus on broad, universal semantic content, often relying on unpaired image-text or text-only corpora. In contrast, unsupervised TBPS emphasizes fine-grained, individual-specific categories using image-only corpora, posing compatibility challenges.

## 2.3 Image Captioning

Image captioning [77] is a vision-language task that aims to generate accurate and fluent natural language descriptions of images. Early approaches [78] relied on simple template-based sentence generation, dependent on object detectors or attribute predictors. Subsequent research [79], [80] transitioned to leveraging Convolutional Neural Networks (CNNs) to encode visual features, which were then decoded into coherent textual descriptions using Recurrent Neural Networks (RNNs). Recently, Transformer-based architectures [81], [82], [83], [84] have emerged as the dominant approach, being applied both in the visual encoding stage and in the language modeling process. Furthermore, the rapid advancements in multimodal large language models (MLLMs) [85], [86], [87] have introduced new paradigms for image captioning [88], [89], [90]. These models utilized pre-trained vision-language representations to generate more coherent, semantically meaningful captions, significantly advancing the integration of vision and language understanding. Person images in TBPS differ from image captioning, focusing on fine-grained, individual-specific features rather than general semantics. Existing image captioning methods are unsuitable for TBPS. We propose a tiered generation framework to enable fine-grained textual descriptions.

## 2.4 Cross-modal Noise-robust Learning

The problem of noisy correspondence in cross-modal learning has garnered increasing attention recently. Huang *et al.* [91] first identified this issue and introduced a noisy correspondence rectifier that employs neural memorization techniques to categorize data into clean and noisy subsets, subsequently correcting misalignments adaptively. Building on this, Han *et al.* [92] proposed a meta-learning purification strategy aimed at suppressing noisy samples, Dang *et al.* [93] developed an information-theoretic framework designed to enhance feature disentanglement in noisy correspondence scenarios. More recently, Shi *et al.* [94] presented a general framework for filtering out noisy correspondences by utilizing priors from pretrained models. Beyond that, in the TBPS domain, several studies have also addressed the challenge of noisy correspondences. Qin *et al.* [8] handled misaligned person image-text pairs through a confident consensus mechanism for noise filtering and a triplet alignment loss to avoid misleading supervision during training. Another tackled the issue of noise originating from weakly matched image-text pairs, where parts of the text are irrelevant to the corresponding image, often due to pairing images with generated pseudo-texts. Gao *et al.* [54] proposed a hybrid patch-channel masking alongside a noise-guided progressive training strategy to boost model robustness against noise. Tan *et al.* [13] introduced a noise-aware masking technique to identify and mask incorrect words based on token similarity. Li *et al.* [1] further contributed to this field by developing an uncertainty-guided filtration module using a Gaussian mixture model to identify and filter noisy samples. The earlier version of this work [12] explored confidence score-based training to prioritize reliable samples based on fixed confidence scores derived during the generation process. Expanding on this concept, we further develop an adaptive confidence-weighted retrieval learning framework to self-adaptively weight samples.

## 3 METHOD

We focus on solving unsupervised TBPS, in which only a collection of person images  $\mathcal{S}_I = \{I_1, I_2, \dots\}$  is provided. In the section, we formally delineate the proposed two-stage solution, generation-then-retrieval (as illustrated in Figure 3). The solution consists of a novel tiered description generation framework (detailed in Section 3.1) that automatically generates synthetic text data  $\mathcal{S}_{T^{syn}} = \{T_1^{syn}, T_2^{syn}, \dots\}$ , and an adaptive confidence-weighted retrieval learning (detailed in Section 3.2) that enables noise-robust training of the retrieval model based on both the original image data  $\mathcal{S}_I$  and the synthetic text data  $\mathcal{S}_{T^{syn}}$ . For simplicity, we next omit symbolic superscripts and subscripts, using  $I$  and  $T$  to represent an image and a text, respectively.

### 3.1 Tiered Description Generation

In unsupervised TBPS, there are no annotated text data paired with image data, making the first stage focus on generating pseudo texts for each person image. However, pre-trained vision-language models [11] fall short in capturing fine-grained, person-specific details, and template-based methods [1], [12], [14], [56] produce limited, single-

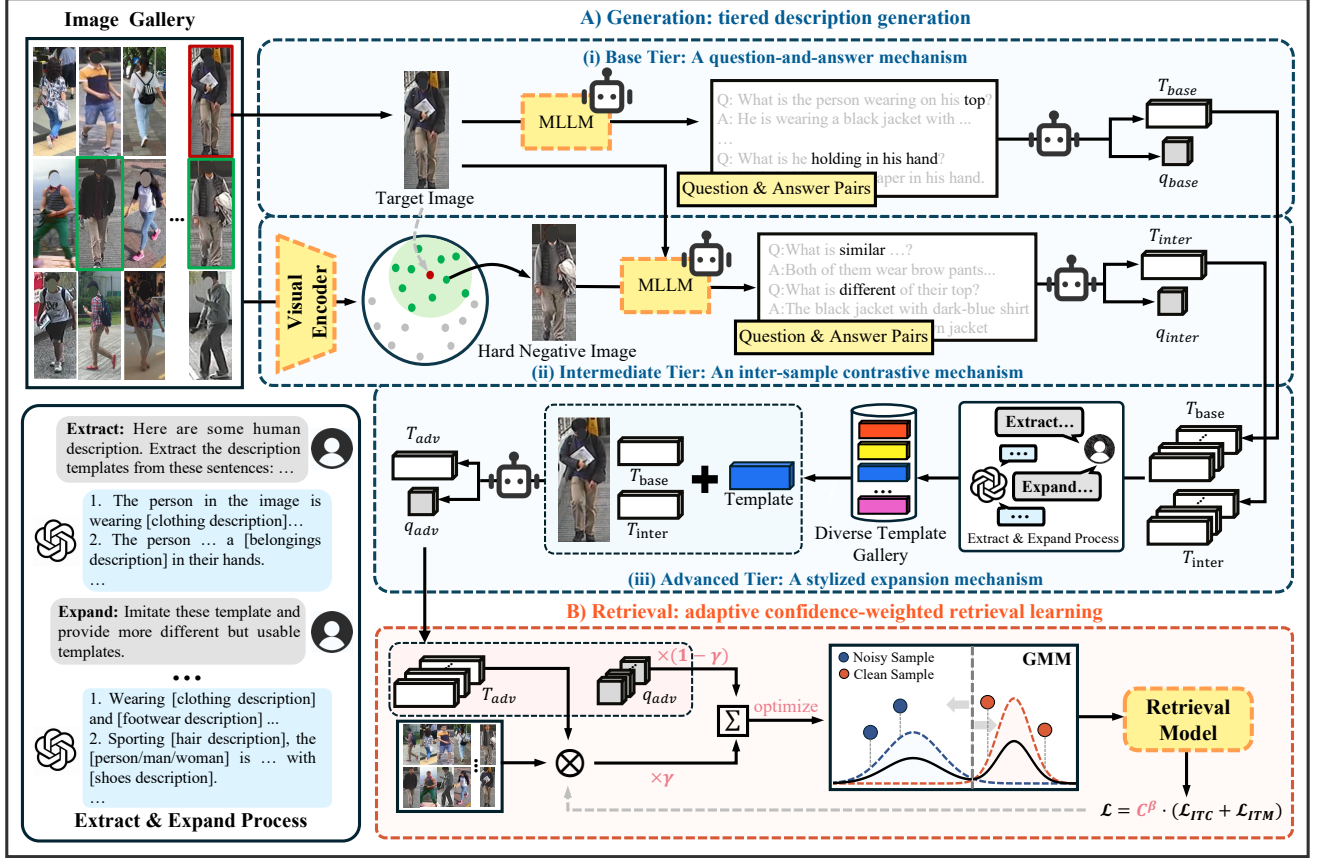


Fig. 3: The overview of the proposed generation-then-retrieval solution (GTR+). The proposed solution consists of a tiered description generation framework and an adaptive confidence-weighted retrieval learning framework. In the first stage, the tiered description generation framework generates fine-grained and diverse textual descriptions through three hierarchical tiers. (i) Base Tier: a question-and-answer mechanism is introduced to generate basic descriptive content; (ii) Intermediate Tier: an inter-sample contrastive mechanism enhances descriptions with finer details; (iii) Advanced Tier: a stylized expansion mechanism diversifies the linguistic styles of the generated text. In the second stage, the adaptive confidence-weighted retrieval learning framework enables noise-robust training by mitigating the impact of noisy textual descriptions generated in the previous stage.

perspective descriptions. To get an enriched description, we propose a three-tiered description generation framework.

### 3.1.1 Base Tier: Question-and-Answer Mechanism

In this tier, we target to generate a basic textual description that captures the essential appearance attributes of the person. Rather than the previous version of this work [12] that rely on template-based generation, we leverage an MLLM to generate the basic textual description through a self-question-and-answer approach.

Specifically, as illustrated in Figure 4, given an input image, we design a structured instruction prompt to guide the MLLM in generating a set of question-and-answer pairs. These pairs probe explicit attribute information of the person, covering global characteristics (e.g., gender, age) and fine-grained details (e.g., shoe color, accessories). This approach ensures the model performs exhaustive yet grounded attribute queries and responses without requiring manual intervention. Subsequently, we apply a sum-instruction to guide the MLLM in extracting and summarizing the information from these question-and-answer pairs, thereby generating a basic textual description that naturally captures the key visual attributes of the image.

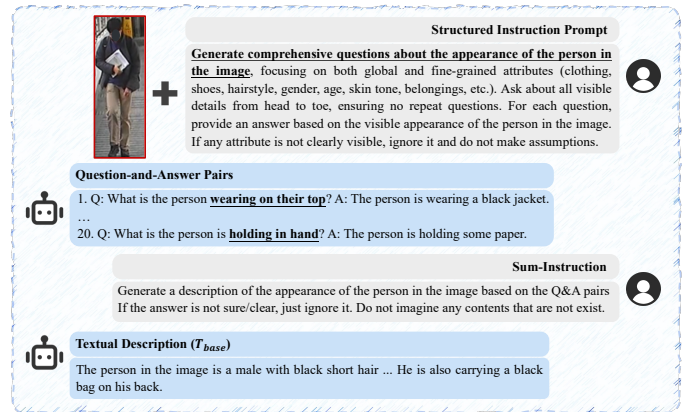


Fig. 4: Flowchart of the question-and-answer mechanism.

This mechanism enables an end-to-end automated approach for extracting pedestrian attributes and generating textual descriptions, eliminating the need for manual intervention and significantly enhancing generalization across diverse images.

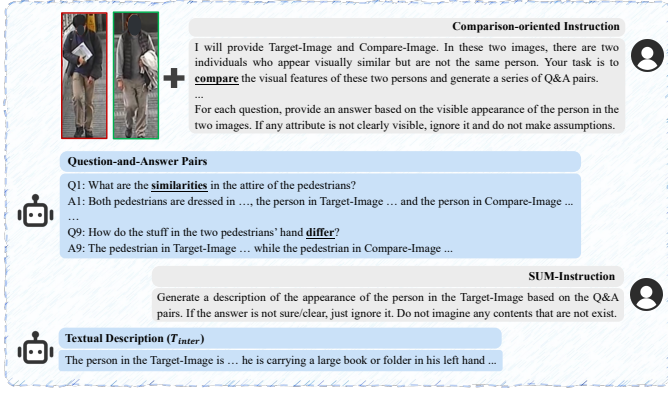


Fig. 5: Flowchart of the inter-sample contrastive mechanism.

### 3.1.2 Intermediate Tier: Inter-Sample Contrastive Mechanism

The base tier generates descriptions containing only basic attribute information, and the intermediate tier improves both granularity and informativeness by incorporating hard negative samples. The MLLM is prompted to compare the target image with the hard negative image, emphasizing similarities and differences. This process enriches the description with finer details and deeper contextual information, improving its quality.

Specifically, we first utilize the DINO-v2 [95] to extract the visual feature  $f^{img}$  of each image in  $S_I$ . Next, we compute pairwise similarity between images using cosine similarity,

$$s(f_i^{img}, f_j^{img}) = \frac{f_i^{img} \cdot f_j^{img}}{\|f_i^{img}\| \cdot \|f_j^{img}\|}, \quad (1)$$

where  $f_i^{img}$  and  $f_j^{img}$  denote the feature of  $i$ -th and  $j$ -th ( $i \neq j$ ) images, respectively. For the target image, we identify its top-10 nearest neighbors based on pairwise similarity and randomly select one as a hard negative image. Instead of always selecting the top-1 nearest neighbor as the hard negative, the random selection from the top-10 nearest neighbors ensures greater diversity in the negative samples. The picked hard negative, while visually similar to the target, is semantically distinct, providing a challenging yet informative contrastive example.

Given the target image and its paired hard negative image, as shown in Figure 5, we enhance the instruction prompt from the base tier by introducing a comparison-oriented instruction. This enhanced instruction guides the MLLM to generate a set of question-and-answer pairs that highlight both shared visual elements (e.g., 'both subjects wear striped shirts') and distinctive visual features (e.g., 'distinct shoe colors: white vs. black') between two images. Subsequently, we apply a revised SUM-Instruction, specifically tailored for the comparison phase, to the MLLM for generating the final text description. This revised SUM-Instruction ensures that the generated caption remains focused solely on the target image while being grounded in the comparative context provided by the hard negative image.

### 3.1.3 Advanced Tier: Stylized Expansion Mechanism

Through the aforementioned two tiers, we obtain textual descriptions that capture essential appearance attributes and fine-grained details of the person images. However, we observe that MLLMs tend to generate captions with limited sentence structures when provided with identical instruction prompts. This structural uniformity in the descriptions could result in overfitting to specific linguistic patterns, thereby diminishing the model's ability to generalize effectively in real-world applications. To address this, we consider to explicitly diversify the styles of textual descriptions in this tier.

Specifically, we begin by randomly sampling a subset of descriptions obtained from the aforementioned two tiers and using ChatGPT-4o to analyze their underlying syntactic templates. Through multiple rounds of interaction with ChatGPT-4o, we iteratively refine these templates to create a diverse array of sentence patterns, which are incorporated into a diverse template gallery.

During textual description generation, a template is randomly selected from the diverse template gallery. The MLLM is then prompted to follow the chosen template and generate a detailed textual description (denoted as  $T_{adv}$ ) using the following instruction,

Generate a detailed description of the person's overall appearance based on the  $T_{base}$  and  $T_{inter}$ , using the following template: {template}. Do not invent or assume any details that are not evident in the image.

Here,  $T_{base}$  and  $T_{inter}$  denote the texts generated by the base tier and the intermediate tier, respectively, and {template} is placeholder that is replaced with the chosen template.

Finally, for each image,  $T_{adv}$  is generated as the paired text. Since  $T_{adv}$  already incorporates descriptive information from both  $T_{base}$  and  $T_{inter}$ , we use only  $T_{adv}$  in constructing the final image-text pair. To further enhance the diversity of the generated texts, we leverage two state-of-the-art, publicly available MLLMs: Qwen2-VL-7B-Instruct [96] and InternVL2.5-8B [97] for generation, respectively. As a result, each image is paired with two texts, forming the final training dataset.

## 3.2 Adaptive Confidence-Weighted Retrieval Learning

The generative stage produces synthetic image-text pairs that, in principle, could serve as supervised signals for training the retrieval model. However, the inherent propensity of MLLMs to introduce hallucinations and inaccuracies during generation leads to noise, thus hindering downstream retrieval performance. To mitigate the effects of such noise, we develop an adaptive confidence-weighted retrieval learning framework.

### 3.2.1 Baseline

We first adopt BLIP [11] as the baseline retrieval model, and yet it can be replaced with others (as verified experimentally in Section 4.6.2) due to the flexible nature of the proposed retrieval learning framework.

BLIP consists of an image encoder  $\mathbf{E}_{img}$ , a text encoder  $\mathbf{E}_{txt}$ , and an image-grounded text encoder  $\mathbf{E}_{i\&t}$ . Given an image  $I = \{p_1, p_2, \dots\}$  and a generated text  $T = \{w_1, w_2, \dots\}$ , where  $p_i$  is  $i$ -th non-overlapping patch and  $w_j$  is  $j$ -th token, we compute the global feature of image, the global feature of text, and the multimodal feature of the image-text pair as,

$$f^{img} = \mathbf{E}_{img}(I), \quad f^{txt} = \mathbf{E}_{txt}(T), \quad (2)$$

$$f^{i\&t} = \mathbf{E}_{i\&t}(\mathbf{E}_{img}(I), \mathbf{E}_{txt}(T)). \quad (3)$$

Then, the optimization objectives with an image-text contrastive learning (ITC) loss and an image-text matching (ITM) loss are computed as,

$$\begin{aligned} L_{itc}^{i2t} &= -E_{(I,T) \sim D} \log \frac{\exp(s(f^{img}, f^{txt})/\tau)}{\sum_{m=1}^M \exp(s(f^{img}, f_m^{txt})/\tau)}, \\ L_{itc}^{t2i} &= -E_{(I,T) \sim D} \log \frac{\exp(s(f^{img}, f^{txt})/\tau)}{\sum_{m=1}^M \exp(s(f_m^{img}, f^{txt})/\tau)}, \\ L_{itc} &= (L_{itc}^{i2t} + L_{itc}^{t2i}) / 2, \end{aligned} \quad (4)$$

where  $M$  is the number of instances in a mini-batch,  $s(f^{img}, f^{txt})$  measures the cosine similarity between the image and the text,  $\tau$  is a learnable temperature;

$$L_{itm} = \mathbb{E}_{(I,T) \sim D} \mathcal{H}(y, \phi(f^{i\&t})), \quad (5)$$

where  $\mathcal{H}$  denotes the cross-entropy function, and  $y$  is 2-dimension one-hot vector and denotes the ground-truth label.  $\phi(f^{i\&t})$  is the predicted matching probability of the pair and  $\phi(\cdot)$  is a linear classification layer.

### 3.2.2 Adaptive Confidence Weighting for Noise-Robust Retrieval

In the baseline retrieval model, the training objectives (Eq. 4 and Eq. 5) treat all image-text pairs equally, assigning uniform weights during optimization. However, noise in the generated texts can mislead training and degrade performance. To address this, we introduce adaptive confidence weighting into the objectives, assigning higher weights to clean samples and lower weights to noisy ones, thereby mitigating the impact of noise.

The confidence score reflects the likelihood of the pair being clean, with high value indicating clean data and vice versa. To estimate the score, we model the similarity distribution of image-text pairs in each training epoch. Based on this distribution, we compute the probability that each pair belongs to the clean data distribution, which is then used as the confidence score for weighting in the subsequent training step.

Specifically, we employ the Gaussian Mixture Model (GMM), which is well known for its ability to capture complex distributions by combining multiple Gaussian components, to model the similarity distribution of image-text pairs,

$$p(\hat{s}) = \sum_{k=1}^K \alpha_k \mathcal{N}(\hat{s} | \mu_k, \sigma_k^2). \quad (6)$$

Here, we set  $K = 2$ , corresponding to two underlying distributions: one representing clean data and the other representing noisy data. The parameter  $\alpha_k$  denotes the

mixing weight of the  $k$ -th Gaussian component, satisfying  $\alpha_1 + \alpha_2 = 1$  and  $\alpha_k \geq 0$  for all  $k$ . The function  $\mathcal{N}(\hat{s} | \mu_k, \sigma_k^2)$  denotes the probability density function of a univariate Gaussian, where  $\mu_k$  and  $\sigma_k^2$  are the mean and variance of the  $k$ -th component, respectively. And  $\hat{s}$  represents the positive pair similarity and is computed as,

$$\hat{s}(f^{img}, f^{txt}) = \gamma \cdot s(f^{img}, f^{txt}) + (1 - \gamma) \cdot q^{txt}, \quad (7)$$

where the cosine similarity  $s(f^{img}, f^{txt})$  is rectified by integrating the text generation probability  $q^{txt}$  in  $\gamma$ -weighted manner. Specifically, during the previous tiered description generation stage, token-level generation probabilities can outputted from the MLLM. Based on them, the text generation probability  $q^{txt}$  is calculated as the mathematical expectation of the generation probabilities of its constituent tokens, serving as a measure of central tendency:

$$q^{txt} = \sum_i^N \frac{1}{N} q_i^{tok}, \quad (8)$$

where  $q_i^{tok}$  is  $i$ -th token generation probability and  $N$  is the number of tokens composing the text. A lower value of  $q^{txt}$  indicates higher uncertainty in the correspondence between the generated text and the associated image, implying potential noise. Notably, the previous version of this work [12] computed the confidence score as the joint probability of token-level scores, which often results in vanishing values for longer captions and thus limited effectiveness. In contrast, our expectation-based confidence measure offers a more objective and robust assessment of text confidence.

The Gaussian Mixture Model in Eq. 6 is optimized using the Expectation-Maximization (EM) algorithm, based on which we compute the posterior probability that a positive pair belongs to the clean data distribution as its confidence score,

$$C = \frac{\alpha_{k^*} \mathcal{N}(\hat{s} | \mu_{k^*}, \sigma_{k^*}^2)}{\sum_{k=1}^2 \alpha_k \mathcal{N}(\hat{s} | \mu_k, \sigma_k^2)}, \quad k^* = \arg \max_k \mu_k. \quad (9)$$

Here, the component with the larger mean,  $\mu_{k^*}$ , represents the clean data distribution.

Finally, we incorporate the confidence score into the optimization objectives as,

$$\begin{aligned} L_{itc}^{i2t} &= -E_{(I,T) \sim D} C^\beta \log \frac{\exp(s(f^{img}, f^{txt})/\tau)}{\sum_{m=1}^M \exp(s(f^{img}, f_m^{txt})/\tau)}, \\ L_{itc}^{t2i} &= -E_{(I,T) \sim D} C^\beta \log \frac{\exp(s(f^{img}, f^{txt})/\tau)}{\sum_{m=1}^M \exp(s(f_m^{img}, f^{txt})/\tau)}, \\ L_{itc} &= (L_{itc}^{i2t} + L_{itc}^{t2i}) / 2, \end{aligned} \quad (10)$$

$$L_{itm} = \mathbb{E}_{(I,T) \sim D} C^\beta \mathcal{H}(y, \phi(\hat{f}^{i\&t})), \quad (11)$$

where  $\beta$  is a hyper-parameter to control the importance of the confidence score.

During each training batch, the model with confidence-weighted objectives updates the similarity distributions and returns the dynamically updated parameters of the GMM, allowing confidence scores to adapt accordingly. These scores are then incorporated into the training objectives. This alternating iterative process ensures that confidence estimation remains sensitive to the current training dynamics, yielding more accurate assessments of data noise compared



Fig. 6: Illustration of the proposed LargeFine-Person dataset and the screened noise images during dataset construction. To ensure data diversity and scale, we generate three textual descriptions for each image, corresponding to the three tiers of our tiered description generation framework, which capture different aspects of person appearance and attributes.

to using a fixed text generation probability  $q^{txt}$  as the confidence score—as done in prior work [12]. Moreover, due to Eq. 7,  $q^{txt}$  provides relatively reliable guidance in the early stages of training, when the model is still unstable and the cosine similarity  $s(f^{img}, f^{txt})$  may be unreliable. As training progresses and features become more discriminative, the cosine similarity increasingly captures meaningful semantic relationships. By jointly modeling both signals within a unified framework, the  $\beta$ -weighted addition ensures robustness to noisy data across different training phases.

### 3.3 Pre-training Dataset Construction: LargeFine-Person

In addition, leveraging the proposed tiered description generation framework, we introduce LargeFine-Person, a large-scale and fine-grained TBPS dataset. It is specifically designed to serve as a generalizable pre-training benchmark for TBPS in unsupervised setting.

**Image Collection.** We adopt LUPerson [17] and LPW [18] datasets as the image sources. LUPerson contains 4 million (M) person images cropped from YouTube videos, while LPW includes 0.59M person images captured by multi-camera systems. Together, these datasets provide a rich variety of environments. However, since the images in LUPerson are extracted from videos using algorithms, they inevitably contain a substantial amount of noise. Thus, we perform data denoising on LUPerson to filter out low-quality images based on the following criteria:

- Low-resolution images: Images with a width smaller than 32 pixels or a height smaller than 64 pixels are

filtered out.

- Blurred images: We use the variance of the Laplacian response to measure image sharpness. Images with a Laplacian variance smaller than 80 are regarded as blurry and filtered out.
- Abnormal person instances: We utilize YOLOv8 for person detection. Images with no detection box or more than one detection box are filtered out.

As a result, we obtain 0.61M high-quality images from the LUPerson dataset, which, combined with the 0.59M images from the LPW dataset, yield a total of 1.2M images.

**Text Generation.** Using the tiered description generation framework, we generate textual descriptions for each of the 1.2M images, incorporating all generated texts ( $T_{base}$ ,  $T_{inter}$ , and  $T_{adv}$ ) to enhance diversity. To maintain generation efficiency, we randomly utilize either Qwen2-VL-7B-Instruct [96] or InternVL2.5-8B [97] within this framework. Finally, this process yields three textual descriptions per image, resulting in a total of 3.6M image-text pairs.

In summary, the proposed tiered description generation framework enables the construction of LargeFine-Person entirely without manual annotation. The characteristics of LargeFine-Person are summarized in Table 1, and example illustrations are shown in Figure 6.

## 4 EXPERIMENT

### 4.1 Datasets

Our experiments involve three open benchmark datasets and one proposed pre-training dataset.

| Methods                | Reference | CUHK-PEDES |       |       |       | ICFG-PEDES |       |       |       | RSTPReid |       |       |       |
|------------------------|-----------|------------|-------|-------|-------|------------|-------|-------|-------|----------|-------|-------|-------|
|                        |           | R@1        | R@5   | R@10  | mAP   | R@1        | R@5   | R@10  | mAP   | R@1      | R@5   | R@10  | mAP   |
| Supervised TBPS        |           |            |       |       |       |            |       |       |       |          |       |       |       |
| Dual Path [23]         | TOMM-20   | 44.40      | 66.26 | 75.07 | -     | 38.99      | 59.44 | 68.41 | -     | -        | -     | -     | -     |
| ViTAA [30]             | ECCV-20   | 55.97      | 75.84 | 83.52 | 51.60 | 50.98      | 68.79 | 75.78 | -     | -        | -     | -     | -     |
| CMKA [98]              | TIP-20    | 54.69      | 73.65 | 81.86 | -     | 46.49      | 67.14 | 75.18 | -     | -        | -     | -     | -     |
| LapsCore [40]          | ICCV-21   | 63.40      | -     | 87.80 | -     | -          | -     | -     | -     | -        | -     | -     | -     |
| SRCF [26]              | ECCV-22   | 64.04      | 82.99 | 88.81 | -     | 57.18      | 75.01 | 81.49 | -     | -        | -     | -     | -     |
| IRRA [99]              | CVPR-23   | 73.38      | 89.93 | 93.71 | 66.13 | 63.46      | 80.25 | 85.82 | 38.06 | 60.20    | 81.30 | 88.20 | 47.17 |
| RaSa [2]               | IJCAI-23  | 76.51      | 90.29 | 94.25 | 69.38 | 65.28      | 80.40 | 85.12 | 41.29 | 66.90    | 86.50 | 91.35 | 52.31 |
| MACF [100]             | IJCV-24   | 73.33      | 88.57 | 93.02 | -     | 62.95      | 79.93 | 85.04 | -     | -        | -     | -     | -     |
| TBPS-CLIP [45]         | AAAI-24   | 73.54      | 88.19 | 92.35 | 65.38 | 65.05      | 80.34 | 85.47 | 39.83 | 61.95    | 83.55 | 88.75 | 48.26 |
| AUL* [16]              | AAAI-24   | 77.23      | 90.43 | 94.41 | -     | 69.16      | 83.32 | 88.37 | -     | 71.65    | 87.55 | 92.05 | -     |
| RDE [8]                | CVPR-24   | 75.94      | 90.14 | 94.12 | 67.56 | 67.68      | 82.47 | 87.36 | 40.06 | 65.35    | 83.95 | 89.90 | 50.88 |
| PTMI [38]              | TIFS-25   | 76.02      | 89.93 | 94.14 | 70.85 | 66.54      | 81.50 | 86.50 | 46.05 | 63.35    | 81.95 | 89.10 | 52.46 |
| GAHR [28]              | TIFS-25   | 76.64      | 90.46 | 94.10 | 66.81 | 68.69      | 82.83 | 87.40 | 42.10 | 68.85    | 86.50 | 91.10 | 53.60 |
| ICL [47]               | CVPR-25   | 78.18      | 91.63 | 94.83 | 69.58 | 69.22      | 83.49 | 88.06 | 42.34 | 70.00    | 86.60 | 91.70 | 54.16 |
| Weakly-supervised TBPS |           |            |       |       |       |            |       |       |       |          |       |       |       |
| CMMT [49]              | ICCV-21   | 57.10      | 78.14 | 85.23 | -     | -          | -     | -     | -     | -        | -     | -     | -     |
| CPCL [50]              | arXiv-24  | 70.03      | 87.28 | 91.78 | 63.19 | 62.60      | 79.09 | 84.46 | 36.16 | 58.35    | 81.05 | 87.65 | 45.81 |
| DG-CMIA [53]           | arXiv-25  | 73.06      | 89.21 | 93.44 | 64.88 | 63.71      | 79.90 | 85.38 | 36.87 | 61.30    | 82.00 | 88.10 | 47.20 |
| Semi-supervised TBPS   |           |            |       |       |       |            |       |       |       |          |       |       |       |
| STBPS [54]             | arXiv-24  | 63.87      | 82.20 | 87.70 | 57.18 | 46.46      | 64.34 | 71.60 | 26.90 | 56.45    | 78.95 | 87.05 | 44.45 |
| Unsupervised TBPS      |           |            |       |       |       |            |       |       |       |          |       |       |       |
| GTR [12]               | ACMMM-23  | 47.53      | 68.23 | 75.91 | 42.91 | 28.25      | 45.21 | 53.51 | 13.82 | 45.60    | 70.35 | 79.95 | 33.30 |
| GAAP [56]              | IJCAI-24  | 47.64      | 67.79 | 76.08 | 41.28 | 27.12      | 44.91 | 53.56 | 11.43 | 44.45    | 65.15 | 75.30 | 31.21 |
| MUMA [1]               | AAAI-25   | 59.52      | 77.79 | 84.65 | 52.75 | 38.11      | 56.01 | 63.96 | 19.02 | 54.35    | 76.05 | 83.65 | 40.50 |
| GTR+                   | Ours      | 61.35      | 79.35 | 85.75 | 55.75 | 47.81      | 64.97 | 71.94 | 28.75 | 54.75    | 75.15 | 83.50 | 43.79 |
| GTR+(Pre-training)     | Ours      | 64.65      | 80.72 | 86.78 | 58.67 | 52.78      | 67.94 | 73.91 | 33.99 | 55.70    | 76.55 | 84.25 | 43.86 |

TABLE 2: Performance comparison of TBPS methods under different settings. Supervised TBPS uses a large number of images with manually annotated texts for training. Weakly-supervised TBPS only requires image-text pairs, without identity annotations. Semi-supervised TBPS leverages a small set of annotated images (1% of all training data) and a large set of unannotated ones. Unsupervised TBPS operates without any textual annotations. GTR<sup>+</sup>(Pre-training) denotes that our proposed GTR<sup>+</sup> can be pre-trained on our LargeFine-Person dataset.

**CUHK-PEDES** [22] is the most commonly-used dataset in TBPS. It consists of 40, 206 images and 80, 440 texts from 13, 003 identities in total, which are split into 34, 054 images and 68, 126 texts from 11, 003 identities in the training set, 3, 078 images and 6, 158 texts from 1, 000 identities in the validation set, and 3, 074 images and 6, 156 texts from 1, 000 identities in the test set. The average length of all texts is 23.

**ICFG-PEDES** [101] contains 54, 522 images from 4, 102 identities in total. Each of the images is described by one text. The dataset is split into 34, 674 images from 3, 102 identities in the training set, and 19, 848 images from 1, 000 identities in the test set. On average, there are 37 words for each text.

**RSTPReid** [102] consists of 20, 505 images of 4, 101 identities. Each identity has 5 corresponding images captured from different cameras. Each image is annotated with 2 textual descriptions, and each description is no shorter than 23 words. There are 3, 701/200/200 identities utilized for training/validation/testing, respectively.

**LargeFine-Person** is our proposed large-scale, fine-grained description dataset for pre-training TBPS. It comprises 1.2M images covering 611, 460 identities, with each image annotated with 3 textual descriptions on average. The descriptions exhibit rich linguistic variation and fine-grained detail, with an average length of 61 words per description, and up to 507 words in total.

During the experiments, unless otherwise specified, we evaluate the proposed GTR+ on each dataset under an unsupervised setting. Specifically, only the images from the training set are used for training, while the corresponding textual annotations in the training set are excluded during the training process.

## 4.2 Protocol

We adopt the widely-used Rank@K (R@K for short, K=1, 5, 10) metric to evaluate the performance of the proposed method. Specifically, given a query text, we rank all the test images via the similarity with the query text, and the search is deemed to be successful if top-K images contain any corresponding identity. R@K is the percentage of successful searches. In addition, we also adopt the mean average precision (mAP) as a complementary metric. Rank@K reflects the accuracy of the first few retrieval results, while mAP emphasizes the comprehensive performance of the method.

## 4.3 Implementation Details

We train the retrieval model using image inputs resized to  $256 \times 256$  and textual inputs truncated to a maximum of 77 tokens. The model is optimized using the Adam optimizer with a learning rate of  $1e-6$ , a batch size of 25, and trained for 15 epochs. All experiments are conducted on NVIDIA

| Pre-training Datasets   | Baselines | Fine-tuning | CUHK-PEDES   |              |              |              | ICFG-PEDES   |              |              |              | RSTPReid     |              |              |              |
|-------------------------|-----------|-------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|
|                         |           |             | R@1          | R@5          | R@10         | mAP          | R@1          | R@5          | R@10         | mAP          | R@1          | R@5          | R@10         | mAP          |
| LUPerson-T [19]         | RDE [8]   | ×           | 20.32        | 38.09        | 48.38        | 19.14        | 9.10         | 20.64        | 28.24        | 3.86         | 22.20        | 44.55        | 58.15        | 17.03        |
|                         |           | ✓           | 53.51        | 73.80        | 81.69        | 48.23        | 42.44        | 58.98        | 66.33        | 21.88        | 51.80        | 72.80        | 80.95        | 38.75        |
|                         | IRRA [99] | ×           | 21.10        | 38.54        | 48.52        | 19.51        | 10.67        | 21.95        | 28.19        | 4.73         | 22.51        | 45.68        | 59.36        | 17.11        |
|                         |           | ✓           | 57.65        | 75.81        | 82.34        | 51.46        | 41.12        | 59.13        | 67.07        | 21.96        | 49.10        | 72.85        | 81.45        | 37.52        |
|                         | BLIP [11] | ×           | 26.98        | 46.18        | 55.36        | 23.59        | 12.47        | 22.04        | 28.22        | 5.30         | 26.15        | 52.20        | 64.20        | 19.15        |
| LUPerson-MLLM [13]      | RDE [8]   | ×           | 54.39        | 73.02        | 80.34        | 48.48        | 36.41        | 54.99        | 63.49        | 18.84        | 49.80        | 72.45        | 82.00        | 37.05        |
|                         |           | ✓           | 55.88        | 76.32        | 82.94        | 49.73        | 42.80        | 60.20        | 67.61        | 22.88        | 52.80        | 73.80        | 83.00        | 39.29        |
|                         | IRRA [99] | ×           | 57.60        | 75.94        | 82.77        | 51.44        | 38.30        | 56.60        | 64.54        | 20.43        | 51.50        | 73.92        | 82.65        | 37.34        |
|                         |           | ✓           | 60.41        | 79.56        | 85.51        | 54.24        | 46.00        | 62.45        | 69.63        | 24.67        | 54.70        | 75.40        | 84.10        | 40.14        |
|                         | BLIP [11] | ×           | 60.11        | 77.45        | 84.15        | 53.46        | 41.97        | 59.75        | 67.01        | 20.81        | 52.40        | 75.15        | 82.80        | 38.81        |
| LargeFine-Person (Ours) | RDE [8]   | ×           | 56.45        | 75.05        | 82.10        | 50.05        | 41.05        | 59.15        | 66.75        | 20.78        | 50.70        | 73.65        | 82.40        | 37.15        |
|                         |           | ✓           | 56.97        | 75.93        | 82.78        | 50.78        | 44.47        | 60.71        | 67.75        | 23.78        | 50.85        | 72.15        | 81.70        | 38.42        |
|                         | IRRA [99] | ×           | 59.44        | 78.54        | 85.22        | 54.11        | 43.77        | 60.77        | 68.05        | 22.30        | 50.45        | 73.45        | 82.35        | 37.68        |
|                         |           | ✓           | 60.88        | 80.02        | 86.45        | 55.54        | 46.02        | 62.76        | 69.92        | 24.58        | 54.40        | 76.05        | 84.10        | 41.31        |
|                         | BLIP [11] | ×           | 62.65        | 78.80        | 84.76        | 55.27        | 47.53        | 64.32        | 71.39        | 25.38        | 52.00        | 74.05        | 82.35        | 38.72        |
|                         |           | ✓           | <b>64.65</b> | <b>80.72</b> | <b>86.78</b> | <b>58.67</b> | <b>52.78</b> | <b>67.94</b> | <b>73.91</b> | <b>33.99</b> | <b>55.70</b> | <b>76.55</b> | <b>84.25</b> | <b>43.86</b> |

TABLE 3: Comparison of pre-training datasets under the unsupervised setting. Only two pre-training datasets, LUPerson-T and LUPerson-MLLM, conform to the unsupervised setting (see Table 1) and are used for a fair comparison. × Fine-tuning: the baseline is only trained on the pre-training dataset and evaluated directly. ✓ Fine-tuning: the baseline is further fine-tuned on the respective downstream dataset (images from the original dataset paired with synthesized descriptions via our tiered description generation framework). Best and second-best results are denoted in **bold** and underline, respectively.

L40 GPUs. In addition, the hyper-parameter  $\gamma$  in Eq. 7 is set to 0.2, and  $\beta$  in Eq. 10 and Eq. 11 is set to 0.8.

#### 4.4 Comparisons with the State-of-the-Art Methods

We compare the proposed GTR+ with SOTA methods on CUHK-PEDES, ICFG-PEDES and RSTPReID in Table 2. These SOTA methods fall into four distinct learning paradigms. Supervised TBPS relies on fully annotated image-text pairs; weakly supervised TBPS requires image-text pairs without identity labels; semi-supervised TBPS leverages a small annotated set (*i.e.*, 1% of data) and a large amount of unannotated images; unsupervised TBPS operates entirely without textual annotations. Among these settings, the unsupervised paradigm provides the most fair and direct comparison with our GTR+. In contrast, the other paradigms leverage additional supervision signals, which naturally confer a performance advantage. As expected, GTR+ exhibits a performance gap compared to methods that utilize more supervision signals, due to their access to richer annotated data. Whereas, compared to unsupervised methods, GTR+ achieves significantly superior performance, demonstrating its effectiveness. Notably, when further pre-trained on our constructed LargeFine-Person dataset—still under an unsupervised setting—GTR+ achieves even stronger results, even outperforming semi-supervised method. This highlights the potential of our constructed large-scale dataset.

Furthermore, we compare the proposed LargeFine-Person pre-training dataset with other pre-training datasets in terms of performance. To this end, we pre-train various retrieval baselines on each dataset, and the results are presented in Table 3 and Table 4. ① We first conduct comparisons under unsupervised setting. Notably, only LUPerson-T and LUPerson-MLLM are constructed independently of human annotations, thus aligning with the unsupervised paradigm. For a fair comparison, we compare our dataset

|            | Pre-training Datasets   | R@1          | R@5          | R@10         | mAP          |
|------------|-------------------------|--------------|--------------|--------------|--------------|
| CUHK-PEDES | w/ Human-annotation     |              |              |              |              |
|            | MALS [20]               | 74.05        | 89.48        | 93.64        | 66.57        |
|            | SYNTH-PEDES [14]        | 74.88        | 89.58        | 94.29        | 67.10        |
|            | HAM-PEDES [15]          | 77.71        | 91.42        | 94.57        | 69.68        |
|            | w/o Human-annotation    |              |              |              |              |
|            | LUPerson-T [19]         | 74.37        | 89.51        | 93.97        | 66.60        |
|            | LUPerson-MLLM [13]      | 76.82        | 91.16        | 94.46        | 69.55        |
|            | LargeFine-Person (Ours) | <b>77.13</b> | <b>90.82</b> | <b>94.49</b> | <b>68.37</b> |
| ICFG-PEDES | w/ Human-annotation     |              |              |              |              |
|            | MALS [20]               | 64.37        | 80.75        | 86.12        | 38.85        |
|            | SYNTH-PEDES [14]        | 64.31        | 80.92        | 86.53        | 38.49        |
|            | HAM-PEDES [15]          | 68.25        | 83.30        | 88.15        | 42.30        |
|            | w/o Human-annotation    |              |              |              |              |
|            | LUPerson-T [19]         | 64.50        | 80.24        | 85.74        | 38.22        |
|            | LUPerson-MLLM [13]      | 67.05        | 82.16        | 87.33        | 41.51        |
|            | LargeFine-Person (Ours) | <b>67.80</b> | <b>82.81</b> | <b>87.66</b> | <b>41.00</b> |
| RSTPReid   | w/ Human-annotation     |              |              |              |              |
|            | MALS [20]               | 61.90        | 80.60        | 89.30        | 48.08        |
|            | SYNTH-PEDES [14]        | 64.00        | 82.85        | 89.90        | 50.06        |
|            | HAM-PEDES [15]          | 71.69        | 87.85        | 93.30        | 55.19        |
|            | w/o Human-annotation    |              |              |              |              |
|            | LUPerson-T [19]         | 62.20        | 83.30        | 89.75        | 48.33        |
|            | LUPerson-MLLM [13]      | 68.50        | 87.15        | 92.10        | 53.02        |
|            | LargeFine-Person (Ours) | <b>69.05</b> | <b>86.90</b> | <b>92.25</b> | <b>54.19</b> |

TABLE 4: Comparison of pre-training datasets under supervised setting. We adopt IRRA [99] as the retrieval baseline.

against these two pre-training datasets. As shown in Table 3, when the retrieval baseline is trained on the pre-training dataset and evaluated directly (*i.e.*, without fine-tuning), our LargeFine-Person consistently achieves superior performance across all baselines, demonstrating its effectiveness. Furthermore, owing to our proposed tiered description generation framework, downstream datasets—CUHK-PEDES, ICFG-PEDES, and RSTPReid—can be expanded by synthesizing paired textual descriptions using only image data, fully adhering to the unsupervised setting. The baseline models are then fine-tuned on these synthesized datasets

| Methods                   | Base tier |     | Intermediate tier |     | Advanced tier |        | R@1   | R@5   | R@10  | mAP   |
|---------------------------|-----------|-----|-------------------|-----|---------------|--------|-------|-------|-------|-------|
|                           | Temp.     | Q&A | Rand.             | ISC | Intg.         | StyExp |       |       |       |       |
| TierG_Base <sub>T</sub>   | ✓         |     |                   |     |               |        | 47.27 | 67.58 | 75.62 | 42.28 |
| TierG_Base <sub>Q</sub>   |           | ✓   |                   |     |               |        | 54.82 | 73.91 | 80.99 | 49.42 |
| TierG_Interm <sub>R</sub> |           |     | ✓                 |     |               |        | 53.33 | 73.16 | 80.49 | 47.80 |
| TierG_Interm <sub>I</sub> |           |     |                   | ✓   |               |        | 54.58 | 73.85 | 81.03 | 49.31 |
| TierG <sub>I</sub>        |           | ✓   |                   | ✓   | ✓             |        | 55.23 | 74.24 | 81.51 | 49.72 |
| TierG                     |           | ✓   |                   | ✓   |               | ✓      | 55.58 | 74.40 | 81.87 | 50.12 |

TABLE 5: Ablation Study of the proposed tiered description generation framework (TierG) on CUHK-PEDES. Q&A: question-and-answer mechanism; ISC: inter-sample contrastive mechanism; StyExp: stylized expansion mechanism. Temp.: template-based generation (vs. Q&A); Rand.: random negative sampling (vs. ISC); Intg.: direct integration without stylized expansion (vs. StyExp).

|             | R@1          | R@5          | R@10         | mAP          |
|-------------|--------------|--------------|--------------|--------------|
| Baseline    | 55.58        | 74.40        | 81.87        | 50.12        |
| k=1         | 57.94        | 76.92        | 83.77        | 51.66        |
| k=5         | 58.45        | 77.25        | 84.01        | 52.01        |
| k=10 (Ours) | <b>59.50</b> | <b>78.49</b> | <b>84.67</b> | <b>53.56</b> |

TABLE 6: Sensitivity analysis of the top- $k$  choice for hard negative selection in the intermediate tier on CUHK-PEDES. Best results are denoted in bold.

(i.e., with fine-tuning). It can be seen that LargeFine-Person continues to outperform the competing datasets across all baselines in this scenario as well. Moreover, we observe a significant performance gain across all baselines after fine-tuning, which further validates the high-quality of the synthesized downstream dataset and the effectiveness of our tiered description generation framework. ② We further conduct comparisons with other pre-training datasets under the supervised setting. In this setup, the retrieval model is first pre-trained on the pre-training dataset and then conventionally fine-tuned on the downstream dataset. As shown in Table 4, the model pre-trained on our LargeFine-Person achieves competitive performance compared to other unsupervised pre-training datasets (i.e., those constructed without human annotation). However, it is slightly inferior to pre-training datasets that leverage human-annotated data during construction—particularly on the HAM-PEDES dataset. This performance gap is reasonable: HAM-PEDES is generated with reference to human-annotated textual descriptions, which ensures higher semantic fidelity. In contrast, LargeFine-Person is constructed entirely without human supervision. As a result, HAM-PEDES contains higher-quality image-text pairs, leading to stronger pre-training efficacy.

## 4.5 Ablation Study

### 4.5.1 Effectiveness of the tiered description generation framework

We propose a tiered description generation framework (TierG) for textual description generation, which incorporates three key components: the base tier with the question-and-answer mechanism (Q&A), the intermediate tier with the inter-sample contrastive mechanism (ISC), and the advanced tier with the stylized expansion mechanism (StyExp). In this part, we conduct an ablation study to evaluate

the contribution of each component, as summarized in Table 5. ① The goal of the base tier is to generate a basic textual description. In previous work [12], we employ a template-based approach (Temp.), in which the instruction is used to prompt the vision-language model to extract attributes, followed by filling these attributes into a manually designed template to construct the final description. This serves as a comparison for our proposed Q&A mechanism. Compared to Temp., Q&A offers clear performance advantages. Since Q&A provides greater flexibility in attribute generation and eliminates the need for laborious manual template design, enabling more natural and varied descriptions. ② For the intermediate tier, the objective is to generate fine-grained textual descriptions. For this, we introduce hard negative images selected from the top- $k$  most similar images as comparison during the generation process (ISC). In contrast, randomly selecting a negative image from all images (Rand.) is a feasible solution, but leads to the degradation in performance. The use of hard negatives in ISC provides more informative contrastive signals, which significantly benefits the model in generating more precise and discriminative descriptions. ③ In the advanced tier, we aim to generate descriptions with diversified styles. For this, we design an instruction that builds upon the outputs from the first two tiers, guided by an explicitly stylized template to steer the MLLM toward generating stylistically rich text (StyExp). Alternatively, we can omit the stylized template and simply integrate the descriptions from the base and intermediate tiers for generation (Intg.). Results show that StyExp achieves superior performance, demonstrating that explicitly diversifying the style of textual descriptions is effective for improving TBPS performance. ④ Overall, our proposed TierG employs a three-tier sequential architecture to generate high-quality textual descriptions in a progressive manner. Results show that TierG outperforms TierG\_Base<sub>Q</sub> and TierG\_Interm<sub>I</sub>, which strongly validates the effectiveness of the tiered design.

### 4.5.2 Effectiveness of the adaptive confidence-weighted retrieval learning framework

We propose an adaptive confidence-weighted retrieval learning framework (AdaWeight), to mitigate the impact of noisy data introduced during the text generation stage. To evaluate the effectiveness of AdaWeight, we compare with several alternative strategies. Tan et al. [13] introduced LUPerson-MLLM, a large-scale TBPS dataset constructed

| Methods          | Masking     |              |       | Weighting   |                | R@1   | R@5   | R@10  | mAP   |
|------------------|-------------|--------------|-------|-------------|----------------|-------|-------|-------|-------|
|                  | Text (sim.) | Text (prob.) | Image | Fixed conf. | Adaptive conf. |       |       |       |       |
| Baseline         |             |              |       |             |                | 55.58 | 74.40 | 81.87 | 50.12 |
| Sim-TxtMask [13] | ✓           |              |       |             |                | 55.60 | 74.45 | 81.93 | 49.26 |
| Prob-TxtMask     |             | ✓            |       |             |                | 55.67 | 74.51 | 82.13 | 50.76 |
| Rand-ImgMask     |             |              | ✓     |             |                | 56.68 | 75.62 | 82.34 | 51.26 |
| FixWeight [12]   |             |              |       | ✓           |                | 58.72 | 77.71 | 84.28 | 54.10 |
| AdaWeight        |             |              |       |             | ✓              | 61.35 | 79.35 | 85.75 | 55.75 |

TABLE 7: Ablation Study of the proposed adaptive confidence-weighted retrieval learning framework on CUHK-PEDES. Noise-robust learning can be achieved through two main strategies. (1) Masking noisy tokens/patches, including: masking text tokens with low image-text similarity scores (Text (sim.)), masking text tokens with low generation probabilities (Text (prob.)), randomly masking image patches (Image); (2) Weighting training samples based on noise levels, including: Fixed confidence weighting (Fixed conf.), Adaptive confidence weighting (Adaptive conf.).

|              | R@1          | R@5          | R@10         | mAP          |
|--------------|--------------|--------------|--------------|--------------|
| Baseline     | 55.58        | 74.40        | 81.87        | 50.12        |
| $K=1$        | 57.29        | 77.23        | 83.56        | 51.21        |
| $K=2$ (Ours) | <b>61.35</b> | <b>79.35</b> | <b>85.75</b> | <b>55.75</b> |
| $K=3$        | 59.31        | 78.46        | 84.84        | 53.54        |
| $K=4$        | 59.49        | 78.40        | 84.68        | 53.79        |
| $K=5$        | 59.34        | 78.62        | 84.78        | 53.61        |

TABLE 8: Sensitivity analysis of the number of Gaussian components in GMM-based confidence estimation on CUHK-PEDES.  $K$  denotes the number of Gaussian components. Best results are denoted in bold.

using synthetic textual descriptions, and proposed a noise-robust retrieval method by masking text tokens with low image-text similarity scores. Inspired by this work, we observe that besides the weighting strategy, the masking strategy can also serve as an effective means for noise-robust learning. We present the comparison results in Table 7. ① Masking strategy for noise-robust learning. We evaluate three masking approaches: (i) Following [13], we mask text tokens with low similarity scores relative to the paired images (Sim-TxtMask); (ii) Alternatively, we mask tokens with low generation probabilities from the generation model (Prob-TxtMask); (iii) We extend this idea to the image domain by randomly masking image patches (Rand-ImgMask), which can implicitly reduce the influence of noisy correspondences. From Table 7, Rand-ImgMask achieves the best performance, showing a slight improvement over text-based masking variants. The image typically exhibits more semantic coherence than the text and retains more meaningful information after partial masking, enabling more robust cross-modal alignment under noisy conditions. ② Weighting strategy for noise-robust learning. In previous work [12], a fixed confidence-weighting approach (FixWeight) is proposed based on text generation probabilities: higher probabilities indicate more confident generations, implying less noise, and thus greater weights. However, FixWeight fails to capture dynamic changes during training, potentially leading to suboptimal handling of noisy samples. In contrast, our proposed adaptive confidence-weighted approach (AdaWeight) dynamically adjusts sample importance according to both generation confidence and training progress, offering a more principled and flexible solution. ③ Masking vs. Weighting. Overall, the weighting

strategy demonstrates superior effectiveness compared to the masking strategy, with our proposed AdaWeight achieving the best performance and offering clear advantages.

#### 4.5.3 Hyper-parameter analysis

① **Top- $k$  hard negative selection.** We evaluate the sensitivity of the top- $k$  hard negative selection during the intermediate generation tier, as shown in Table 6. While all settings outperform the baseline,  $k = 10$  yields the best results. An overly small  $k$  (e.g.,  $k = 1$ ) restricts candidates to a narrow neighborhood, limiting comparison diversity. Conversely, sampling from the top-10 neighbors strikes an optimal balance: it preserves visual similarity while providing diverse contrastive cues, which generates richer discriminative descriptions and improves retrieval performance. ② **Gaussian components ( $K$ ) and GMM discriminative capability.** To explicitly validate the discriminative capability of our GMM, Figure 7 visualizes its stage-wise behavior. The model consistently separates training pairs into clean and noisy subsets. More importantly, the similarity gap between these two subsets gradually widens as training proceeds. This dynamic confirms that the GMM is not merely fitting a static bimodal shape, but progressively and reliably identifying clean samples as the feature representa-

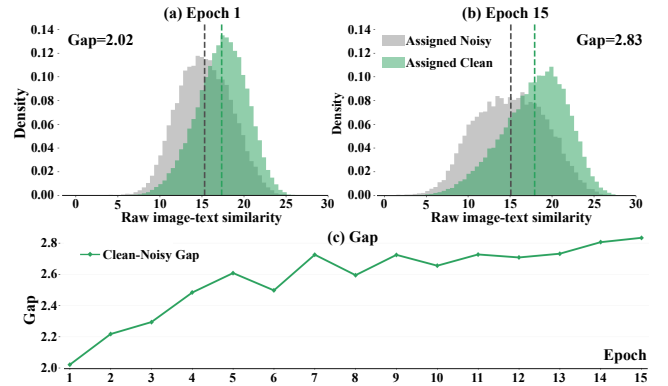
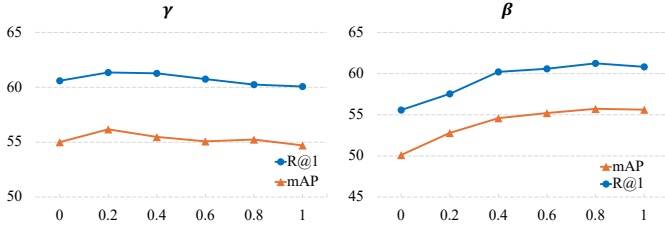


Fig. 7: Visualization of GMM-based separation during training. Panels (a) and (b) show the distributions of image-text similarity for the subsets assigned by the GMM to the noisy and clean components at epochs 1 and 15, respectively. Panel (c) shows the gap between the clean and noisy subsets over training epochs.

Fig. 8: Hyper-parameter analysis of  $\gamma$  and  $\beta$ .

tions improve. Furthermore, we justify our two-component design by analyzing the sensitivity of the Gaussian component number ( $K$ ), as reported in Table. 8. The results demonstrate that  $K = 2$  yields the optimal performance. A single component ( $K = 1$ ) is too coarse to model the underlying noise variation, whereas using more components ( $K > 2$ ) tends to over-partition the distribution, leading to degraded confidence estimation. **Impact of  $\gamma$  and  $\beta$ .** Our adaptive confidence-weighted retrieval learning framework involves two hyper-parameters:  $\gamma$  and  $\beta$ . The parameter  $\gamma$  balances the contributions of pair-wise cosine similarity and text generation probability in computing the confidence scores, while  $\beta$  controls the influence of these confidence scores within the training objectives. A higher  $\gamma$  places more weight on cosine similarity, whereas a lower  $\gamma$  emphasizes text generation probability. And  $\gamma = 1$  corresponds to adopting only cosine similarity for computing the confidence scores, and  $\gamma = 0$  corresponds to using only text generation probability. As shown in Figure 8, relying solely on either component ( $\gamma = 0$  or  $\gamma = 1$ ) leads to suboptimal performance. The best results are achieved by combining both with  $\gamma = 0.2$ . The cosine similarity is an optimizable dynamic value that improves during training and a smaller  $\gamma$  helps maintain training stability. The parameter  $\beta$  governs the overall impact of confidence scores on training. When  $\beta = 0$ , confidence scores have no effect. As illustrated in Figure 8, performance is relatively low when  $\beta < 0.5$ , becomes more stable for  $\beta > 0.5$ , and reaches its peak at  $\beta = 0.8$ .

## 4.6 Generalization Experiment

### 4.6.1 Domain generalization

We evaluate the domain generalization capability of the proposed adaptive confidence-weighted learning framework in Table 9. Specifically, models are trained on the source dataset and directly tested on the target dataset, e.g.,  $C \rightarrow I$ , where training is conducted on CUHK-PEDES and evaluation is performed on ICFG-PEDES. Notably, all compared methods are supervised approaches that rely on both images and human-annotated textual descriptions during training. Moreover, among these, MRA [103] is TBPS method specifically designed for the domain generalization setting. In contrast, our GTR+ operates in an unsupervised manner, requiring only images from the source domain, with textual descriptions automatically generated by our proposed tiered description generation framework. Despite the absence of annotated texts, GTR+ achieves competitive performance, demonstrating the high quality of the syn-

| Methods           |            | R@1   | R@5   | R@10  | mAP   |
|-------------------|------------|-------|-------|-------|-------|
| C $\rightarrow$ I | LGUR [36]  | 34.25 | 52.28 | 60.85 | -     |
|                   | IRRA [99]  | 42.62 | 62.39 | 70.13 | 23.89 |
|                   | BLIP [11]  | 47.93 | 63.98 | 71.07 | 26.25 |
|                   | RDE [8]    | 48.15 | 66.36 | 73.43 | 27.01 |
|                   | GAHR [28]  | 49.00 | 66.34 | 73.59 | -     |
|                   | MRA* [103] | 50.01 | 67.13 | 74.24 | -     |
| GTR+ (Ours)       |            | 48.93 | 65.98 | 71.27 | 28.25 |
| I $\rightarrow$ C | LGUR [36]  | 25.44 | 44.48 | 54.39 | -     |
|                   | IRRA [99]  | 32.21 | 55.12 | 65.19 | 31.72 |
|                   | BLIP [11]  | 36.35 | 57.98 | 67.21 | 34.19 |
|                   | RDE [8]    | 38.33 | 59.51 | 68.30 | 35.56 |
|                   | GAHR [28]  | 40.60 | 62.46 | 71.62 | -     |
|                   | MRA* [103] | 47.66 | 68.70 | 76.82 | -     |
| GTR+ (Ours)       |            | 38.73 | 60.12 | 69.87 | 35.90 |

TABLE 9: Domain generalization comparisons. C: CUHK-PEDES, I: ICFG-PEDES. GTR+ trains in source with an unsupervised setting with generated texts, others rely on fully annotated supervised data. \* denotes the method specifically designed for the domain generalization scenario.

| Baselines | R@1   | R@5   | R@10  | mAP   |
|-----------|-------|-------|-------|-------|
| RDE [8]   | 59.93 | 78.89 | 84.18 | 53.69 |
| IRRA [99] | 60.72 | 79.11 | 84.93 | 54.80 |
| BLIP [11] | 61.35 | 79.35 | 85.75 | 55.75 |

TABLE 10: Results of adopting different retrieval baselines in the adaptive confidence-weighted learning framework on CUHK-PEDES.

thesized descriptions and the effectiveness of our proposed adaptive confidence-weighted retrieval learning framework.

### 4.6.2 Different baseline retrieval models

We evaluate the compatibility of the proposed adaptive confidence-weighted learning framework by integrating various baseline retrieval models. As shown in Table 10, the framework demonstrates strong adaptability and robustness, consistently achieving competitive performance across different retrieval models.

## 4.7 Visualization Analysis

Figure 9 presents diverse examples illustrating the top-5 retrieval results of both the BLIP baseline and our GTR+, demonstrating effectiveness and superiority of GTR+. Figure 10 visualizes the activation maps for positive image-text pairs using BLIP [11], GTR [12] and our GTR+. Despite the unsupervised setting, GTR+ places greater emphasis on the described object in the image and the key attributes in the text during computation, compared to other methods.

## 5 CONCLUSION

In this paper, we address the unsupervised TBPS task, aiming to remove reliance on expensive human-annotated textual descriptions. To this end, we propose GTR+, a two-stage framework that first generates pseudo-descriptions to compensate for the absence of annotations, and subsequently trains the retrieval model in a supervised fashion. In the generation stage, we introduce a tiered description

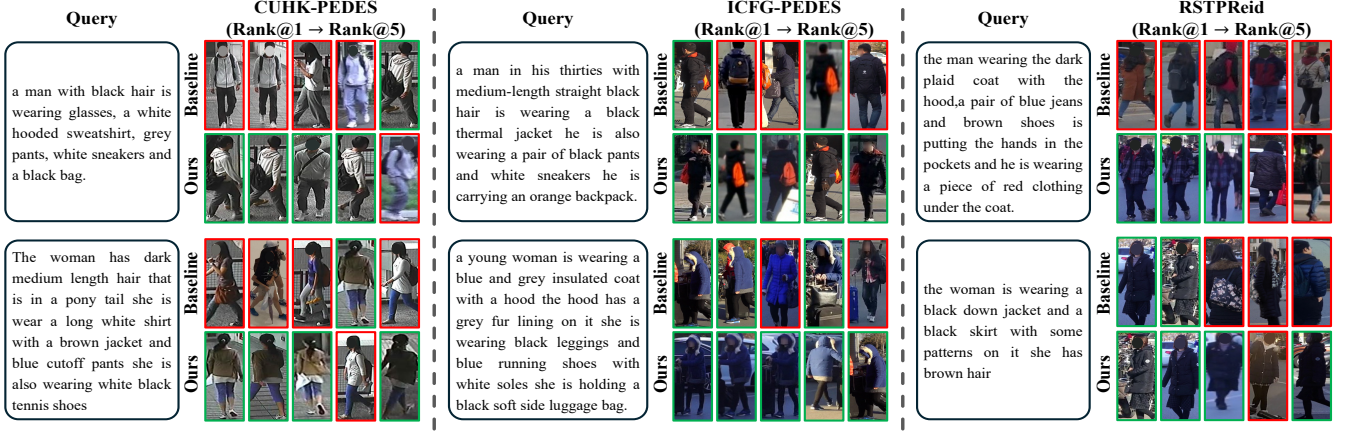


Fig. 9: Visualization of retrieval results (R@1 to R@5). Each row represents a query-response group. Green and red bounding boxes indicate correct and incorrect matches, respectively. BLIP [11] is used as the baseline.

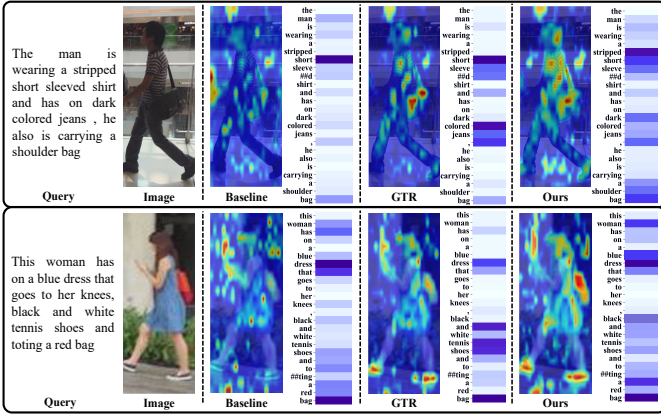


Fig. 10: Visualization of activation maps. Warmer colors indicate stronger activation during model computation. BLIP [11] is used as the baseline.

generation framework that produces fine-grained and stylistically diverse captions through a three-tiered sequential process. In the retrieval stage, we propose an adaptive confidence-weighted learning framework that dynamically adjusts each sample's training contribution based on its confidence score, effectively reducing the negative impact of noisy pseudo texts. Moreover, we contribute LargeFine-Person, a new large-scale dataset featuring high-quality, fine-grained, and diverse textual annotations, which serves as a more realistic and challenging benchmark for TBPS. Extensive experiments demonstrate that GTR+ achieves impressive performance on multiple TBPS benchmarks under the unsupervised setting, and also validate the effectiveness of the proposed dataset.

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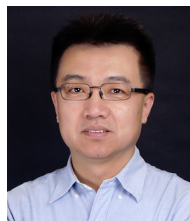


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